Reasoning-Trace Collapse: Evaluating the Loss of Explicit Reasoning During Fine-Tuning

Lukas Twist *Helen Yannakoudakis* *Jie M. Zhang*

SAFE & TRUSTED AI UKRI CENTRE FOR DOCTORAL TRAINING · UK Research and Innovation · King's College London · Imperial College London

Full paper: Lukas Twist:

Motivation

→ Many recent language models produce explicit reasoning.

→ Downstream fine-tuning data often contains instruction-response data with no explicit reasoning.

→ **This creates a mismatch.**

Reasoning Example

Many models use ***<think> tags*** to show their reasoning, for example:

```
<think>
...reasoning...
</think>
Good question, I...
Answer: 42
```

Evaluation Framework

Separate whether a model reasons from whether its answer is correct.

→ Classify the reasoning:

↳ ***Valid*** *- complete reasoning*

↳ ***Empty*** *- nothing inside <think>*

↳ ***Missing*** *- no <think> tags*

↳ ***Truncated*** *- no closing </think>*

→ Track the following metrics:

↳ ***pass@1*** *- final accuracy*

↳ ***VR*** *- valid reasoning rate*

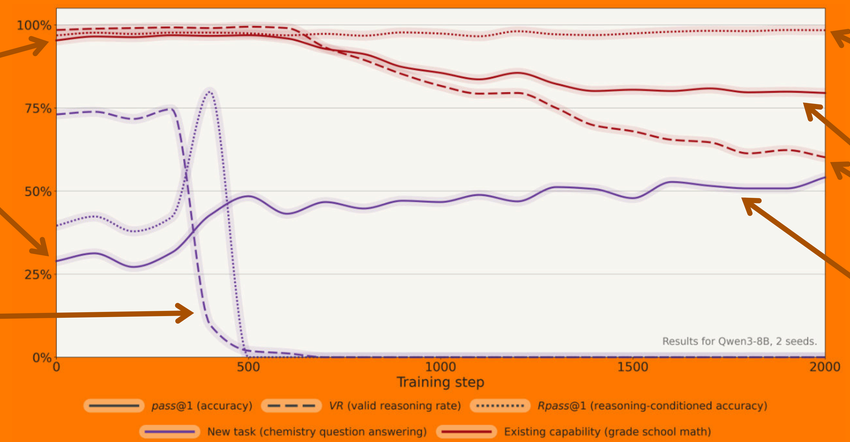
↳ ***Rpass@1*** *- accuracy when the model has valid reasoning*

Accuracy alone makes it look like fine-tuning learns the new task while harming the old one — reasoning-aware metrics tell a different story.

Model starts with very high math ability

Low initial accuracy on chemistry

Model quickly stops producing valid reasoning on chemistry



Math ability stays high when the model reasons

Even though total math accuracy and valid reasoning fall during training

Model gradually learns the new task without producing valid reasoning

Fine-tuning can make reasoning models look worse, but they may just be reasoning less often — remaining strong when valid reasoning is present.

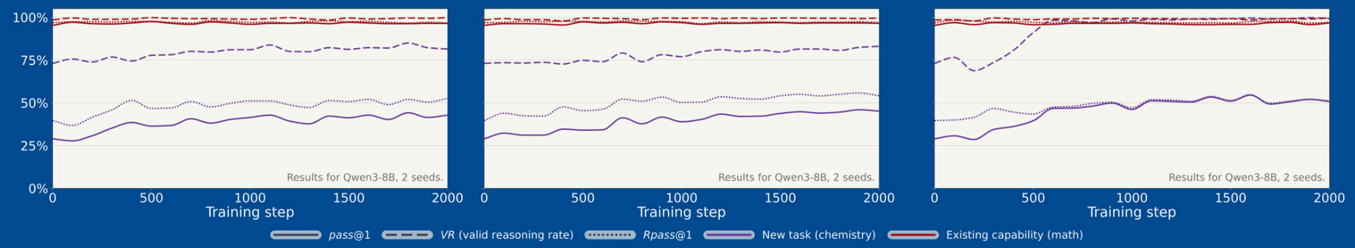
So... can we prevent reasoning-trace collapse?

Loss-masking: Exclude selected token regions from the training loss.
↳ Lightweight and does not require teacher-generated reasoning.

Masked-think: Ignore the loss on the empty reasoning block, so the model is not rewarded for omitting reasoning.

Response-only: Train only on the final-response tokens, masking both the prompt and reasoning region.

Distillation: Generate reasoning with a teacher model and fine-tune on the augmented dataset.
↳ Often effective, but not a guarantee; can be expensive and difficult for private or niche data.



Even lightweight loss-masking can preserve much more reasoning than standard fine-tuning — but effects vary by model.

TAKEAWAYS

1. Fine-tuning can cause models to **stop producing explicit reasoning**.
2. Standard answer accuracy can **misdiagnose what has gone wrong**.
3. Accuracy can remain high **when valid reasoning is present**.
4. **Lightweight loss masking** can sometimes preserve reasoning behaviour.
5. Evaluate **reasoning behaviour alongside final answer accuracy**.

Software Release

We release **ThinkPack**, an open-source library for reasoning-aware training and evaluation:

→ Parse and classify reasoning

→ Compute reasoning-aware metrics

→ Apply reasoning-aware loss masks

```
pip install thinkpack
```

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